

Helicopter Rotor Dynamics: Aerodynamic and Multibody Simulation in MSC ADAMS

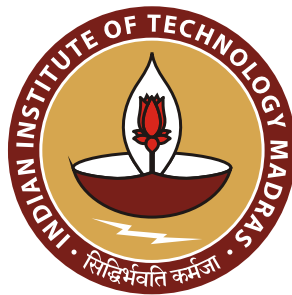
A Project Report

submitted by

MUGDHA MEDA

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for the award of the degree of*

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THESIS CERTIFICATE

This is to certify that the thesis titled **HELICOPTER ROTOR DYNAMICS: AERODYNAMIC AND MULTIBODY SIMULATION IN MSC ADAMS**, submitted by **Mugdha Meda**, to the Indian Institute of Technology, Madras, for the award of the degree of **Bachelors of Technology**, is a bona fide record of the research work done by her under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS: helicopter rotor dynamics; swashplate mechanism; MSC ADAMS;
rotorcraft aerodynamics; rotorcraft CAD-based modeling

Helicopters are a remarkable class of aircraft, capable of vertical takeoff and landing, hovering, and multidirectional flight, which makes them exceptionally versatile for both civilian and military applications. However, these advantages come with significant trade-offs—including complex aerodynamics, increased vibrations, and higher power requirements compared to fixed-wing aircraft of similar weight—largely due to the intricate swashplate-driven rotor system that enables their unique capabilities.

A key aspect of helicopter rotor dynamics is the interplay between aerodynamic forces and blade motion. To accommodate these forces, rotor blades incorporate flapping and lead-lag hinges—either mechanical or designed as flexible "virtual" hinges in modern hubs. These allow the blades to move independently in response to varying lift and drag forces, ensuring stable and controllable flight.

This thesis focuses on studying blade flapping using a mechanically hinged setup with provisions to control the angle of attack (AoA). A CAD model of the swashplate and rotor system is developed, and aerodynamic forces are applied to the blades using MSC ADAMS to analyze the resulting coning angle, which indicates thrust direction. The simulation outcomes are then compared with analytical predictions based on the governing ODEs in MATLAB. This model and validation establishes a foundation for future extensions involving cyclic and collective pitch control and the study of flexible blades with virtual hinge dynamics.

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ABBREVIATIONS

AoA	Angle of Attack
ODE	Ordinary Differential Equation
FFT	Fast Fourier Transform
CAD	Computer-Aided Design
DoF	Degrees of Freedom
COM	Center of Mass
RPM	Revolutions Per Minute
CL	Coefficient of Lift

NOTATIONS

F	Force (N)
L	Lift force (N)
m	Mass of blade (kg)
R	Length of blade (m)
d	Distance between hinge and rotation axis (m)
I_x, I_z	Moment of inertia about hinge axes (kg·m ²)
Ω	Rotational speed (rad/s)
β	Flap angle (rad)
θ	Fixed pitch angle (rad)
θ_c	pitch angle amplitude for cosine harmonic (rad)
w_c	pitch harmonic input frequency (rad/s)
K	Aerodynamic stiffness coefficient (Nm)
C_L	Coefficient of lift
c	Chord length of blade (m)
γ	Lock number (dimensionless)
U_T	Tangential velocity at blade element (m/s)
v_i	Induced velocity (m/s)
ζ	Damping ratio
ω_n	Natural frequency (rad/s)
ω_c	Forcing frequency (rad/s)
β_0	Steady-state flap angle (rad)
β_c	Amplitude of flap angle corresponding to pitch harmonic (rad)

CHAPTER 1

STUDY OF ROTATING BLADE MOTION

Helicopter rotor systems incorporate a swashplate mechanism that enables precise control of blade pitch and allow for blade flapping motions essential to rotor aerodynamics and flight control. Blade pitch is controlled through a pitch bearing that also allows the blades to feather. Similarly, blade flapping is enabled by hinges or flexures at the blade root, allowing each blade to move up and down in response to aerodynamic and inertial forces. This flapping motion helps the blades adjust their angle of attack naturally to balance varying aerodynamic loads during flight.

Cyclic pitch control reorients the rotor thrust vector, tilting the rotor disk in the desired direction and providing the pilot with pitch and roll control of the helicopter, while collective pitch control varies the overall magnitude of the thrust vector.

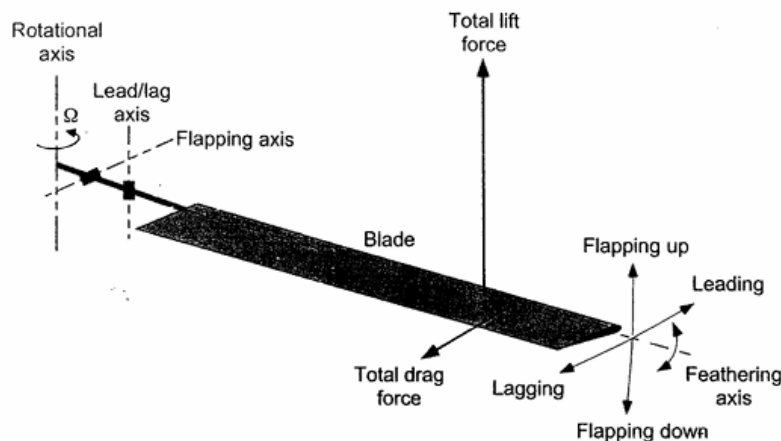


Figure 1.1: Blade degrees of freedom.

In hovering flight, the aerodynamic environment is azimuthally symmetric, meaning each rotor blade experiences the same flow conditions. Under these steady forces, the blades flap up and lag back slightly, reaching an equilibrium position where aerodynamic lift balances centrifugal forces.

The blades “cone” upward with a small coning angle, while aerodynamic drag causes a minor lagging motion. These small motions are governed primarily by

centrifugal forces that maintain rotor stability. This thesis studies blade flapping for hovering flight considering collective pitch applied. The study can further be extended to forward flight with cyclic pitch control.

1.1 Equation of Motion for a Flapping Blade

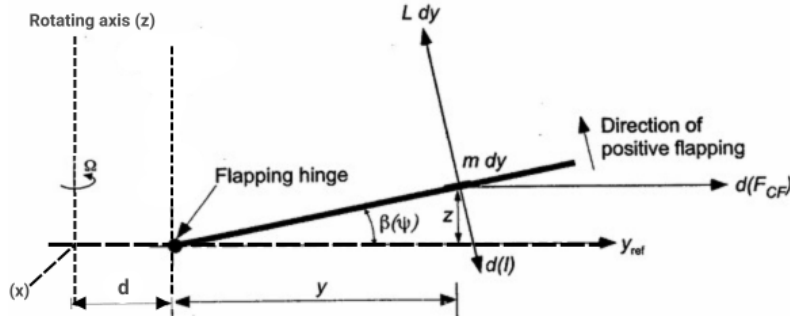


Figure 1.2: Forces acting on an element of a flapping blade.

In hovering flight, the solution for β is a constant ($= \beta_0$) independent of ψ (the azimuthal position $= \Omega t$ for constant speed input) and this angle is called the coning angle. Under forward flight conditions as a result of the cyclically (azimuthally) varying ailerons, the blade flaps up and down in a periodic manner with respect to azimuth. Consider Fig. 2.2, which shows the line of action of the aerodynamic forces, the centrifugal forces, and the inertial forces acting on a small element of the blade span. Define the moment to be positive in a direction such as to reduce β . The centrifugal moment about the hinge is then:

$$d(M_{CF}) = (mdy)(y + d)^2 \Omega^2 \beta = m(y + d)^2 \Omega^2 \beta dy. \quad (1.1)$$

It will be assumed that there is a hinge offset, that is, $d \neq 0$. The inertial moment about the hinge is

$$d(I) = (mdy)y^2 \ddot{\beta} = my^2 \ddot{\beta} dy \quad (1.2)$$

and the aerodynamic moment about the hinge is

$$d(M_\beta) = -Ly dy, \quad (1.3)$$

where the sign of the latter term is noted. Therefore, the equation of motion can be derived by summing the moments about the flap hinge giving

$$\int_0^R d(M_{CF}) + \int_0^R d(I) + \int_0^R d(M_\beta) = 0. \quad (1.4)$$

Introducing the relevant expressions for $d(M_{CF})$, $d(I)$, and $d(M_\beta)$ gives

$$\int_0^R m\Omega^2\beta(y+d)^2 dy + \int_0^R my^2\ddot{\beta} dy - \int_0^R Ly dy = 0. \quad (1.5)$$

Collecting terms results in

$$\left(\int_0^R my^2 dy\right)\ddot{\beta} + \left(\int_0^R m(y+d)^2 dy\right)\Omega^2\beta = \int_0^R Ly dy. \quad (1.6)$$

The mass moment of inertia of the blade about the flap hinge is say

$$I_b = \int_0^R my^2 dy. \quad (1.7)$$

Thus the equation of flapping motion in Eq. 4.29 can be written as approximately

$$I_b\ddot{\beta} + (I_b + m * d^2 * R)\Omega^2\beta = \int_0^R Ly dy. \quad (1.8)$$

Now, consider the aerodynamic forces. At any blade element of chord c at a radial distance y from the rotational axis it can be readily shown using the blade element theory (Section 3.5 in (?)) that with uniform inflow the aerodynamic force per unit length is

$$L = \frac{1}{2}\rho U_T^2 c C_L = \frac{1}{2}\rho U_T^2 c C_{L_0} \left(\theta - \frac{\beta y}{U_T} - \frac{v_i}{U_T} \right), \quad (1.9)$$

so the aerodynamic moment about the hinge will be

$$\int_0^R Ly dy = \frac{1}{2}\rho\Omega^2 c C_L \int_0^R y^3 \left(\theta - \frac{\beta y}{\Omega y} - \frac{v_i}{\Omega y} \right) dy \quad (1.10)$$

$$= \frac{1}{8} \rho \Omega^2 c C_L R^4 \left(\theta - \beta - \frac{4v_i}{3\Omega R} \right) \quad (1.11)$$

Where v_i is induced velocity on flapping: $v_i = \dot{\beta} R$

The Lock number is defined as

$$\gamma = \frac{\rho c C_L R^4}{I_b} \quad (1.12)$$

which can be viewed as a measure of the ratio of aerodynamic forces to inertial forces. For a typical helicopter rotor, the value of the Lock number varies from 5 to 10. Notice, however, that its value depends on the density of the air and so will be affected by changes in density altitude (see Section 5.2 in (?)). The equation governing the behavior of the flapping blade becomes

$$\ddot{\beta} + (1 + (m * d^2 * R)/I_b) \beta = \frac{\rho c C_L R^4}{I_b} \left(\frac{1}{8} \left(\theta - \beta - \frac{4v_i}{3\Omega R} \right) \right), \quad (1.13)$$

which on rearrangement gives roughly

$$\ddot{\beta} + (1 + (m * d^2 * R)/I_b + \frac{\gamma}{8}) \beta = \frac{\gamma}{8} \left(\theta - \frac{\dot{\beta}}{\Omega} \right). \quad (1.14)$$

This is the flapping equation for a *centrally hinged* blade, that is, one that is hinged at the rotational axis with $d \neq 0$. A more general form is to leave the aerodynamic force (moment) on the right-hand side is

$$\ddot{\beta} + (1 + (m * d^2 * R)/I_b) \beta = \gamma M_\beta, \quad (1.15)$$

where

$$M_\beta = K/\gamma * \left(\theta - \frac{\dot{\beta}}{\Omega} \right). \quad (1.16)$$

Notice the similarity of Eq. 1.15 with the equation of motion of a simple degree-of-freedom spring–mass–damper system, that is, $m\ddot{x} + c\dot{x} + kx = F$, where x is the displacement, m is the mass, c is the damping, and k is the spring stiffness. This system has an undamped natural frequency of $\omega_n = \sqrt{k/m}$. Therefore, by

analogy with Eq. 1.15 the natural flapping frequency of the flapping blade about a hinge located at the rotational axis is once per revolution (1/rev) or $\omega_n = \Omega$ rad/s.

Consider first the case where the rotor operates in a vacuum, in which case there are no aerodynamic forces present. The flapping equation reduces to

$$\ddot{\beta} + (1 + (m * d^2 * R)/I_b)\beta = 0. \quad (1.17)$$

It is easy to show that this equation has the general solution

$$\beta = \beta_{nC} \cos \psi + \beta_{nS} \sin \psi, \quad (1.18)$$

where β_{nC} and β_{nS} are arbitrary coefficients. Thus, in the absence of aerodynamic forces, the rotor takes up an arbitrary orientation in inertial space. In effect, the rotor acts like a gyroscope. The introduction of aerodynamic forces produces an aerodynamic flapping moment about the hinge, which causes the rotor to move to an over-coning until the aerodynamic moment (which is contained inside the M_β term) causes equilibrium to be obtained once again.

CHAPTER 2

MODELING AND SIMULATION: BLADE CONSTRAINED TO FLAPPING

To initiate the simulation modeling process, it is essential to construct the system incrementally, validating each stage against corresponding analytical solutions. This thesis begins by modeling a simplified rotor blade rotating, constrained to a single degree of freedom—flapping motion.

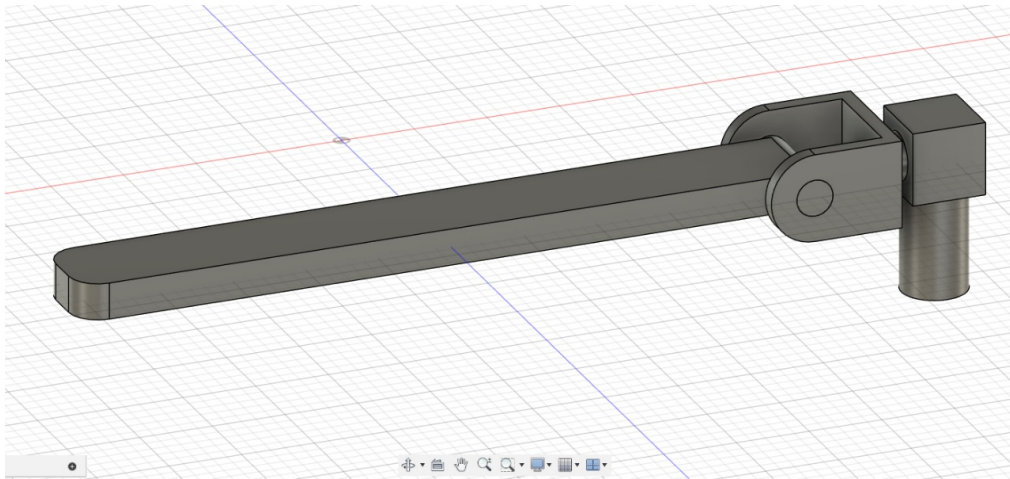


Figure 2.1: Simple Blade Flapping CAD in Autodesk Fusion

The following values were kept consistent with the simulation and analytical formulation:

Parameter	Value	Units
Ω (Motor speed)	25	rad/s
L (Lift)	50	N
m (Mass)	0.287286	Kg
R (Length of blade)	0.377	m
d (Eccentricity of rotating axis with flap hinge)	0.1355	m
I_x (Inertia about flap hinge x)	10	Nm^2
I_z (Inertia about flap hinge z)	10	Nm^2

Table 2.1: Parameters for Basic Flapping Model

A corresponding CAD model as shown in Fig 2.1 was created and imported into MSC ADAMS, where aerodynamic loads were applied in a structured, three-stage verification process to ensure consistency with theoretical predictions.

2.1 Point Lift

The simplest form of applying lift is by applying a constant 'Space Fixed' vertical force for the lift at the center of mass of the blade as shown in Fig 2.2. Then, with respect to Fig 1.2:

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = LR/2. \quad (2.1)$$

Since Lift is acting at the COM

$$M_\beta = LR/2 \quad (2.2)$$

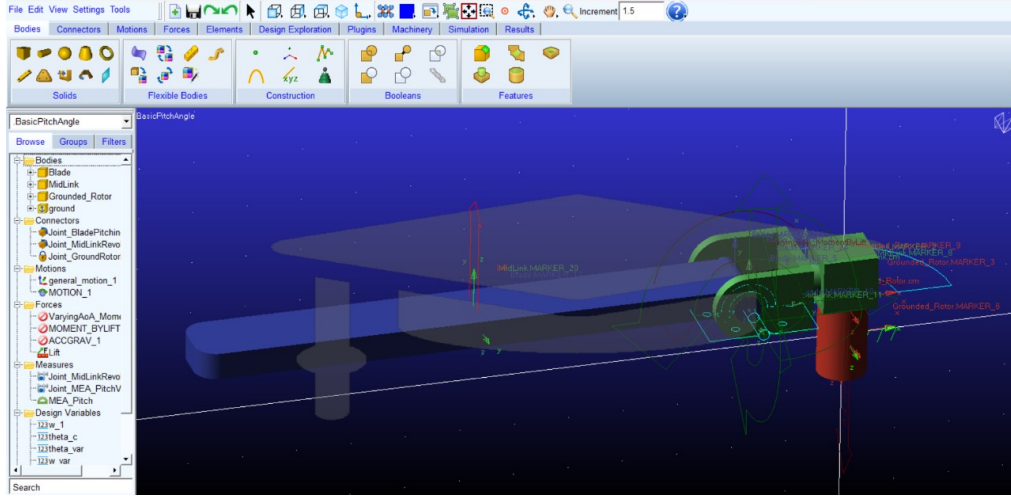


Figure 2.2: Simple Blade Flapping with Point Lift in MSC ADAMS.

The flap angle and velocity was measured in MSC ADAMS and compared with the expected solution from MATLAB. The results from ode45 in MATLAB and MSC ADAMS match to a good degree (10% error in amplitude but small values). The frequency of oscillations match perfectly.

They both oscillate about

$$\beta = M_\beta / (I_x + m * d^2) \quad (2.3)$$

as expected.

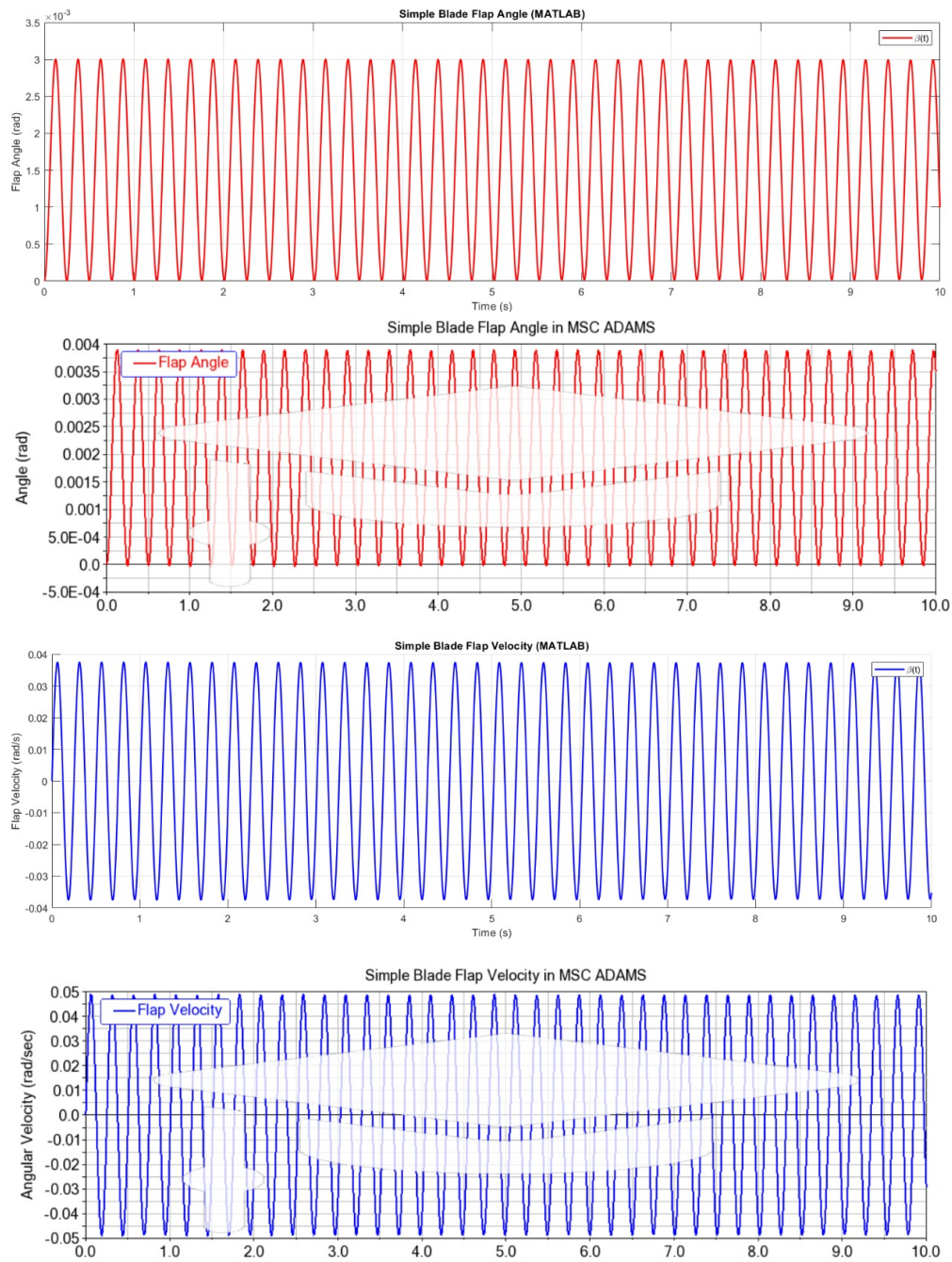


Figure 2.3: Comparison between flap angle and velocity from MATLAB and MSC ADAMS simulations.

2.2 Moment due to Lift with Constant AoA

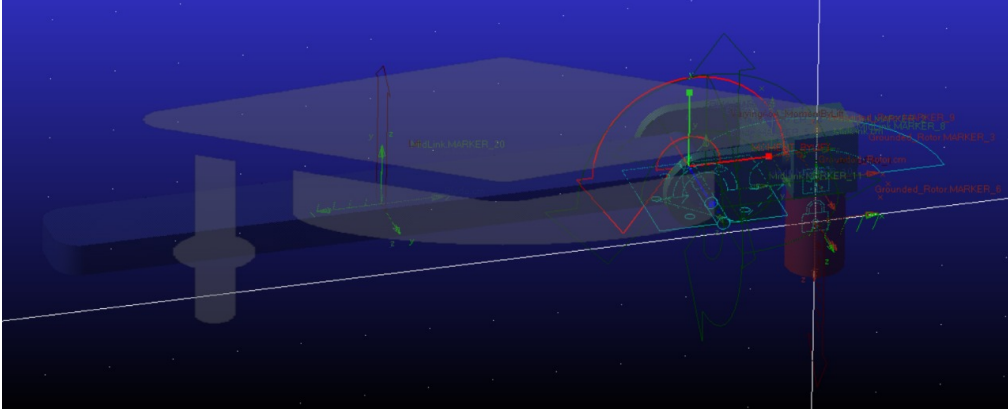


Figure 2.4: Simple Blade Flapping with Moment by Lift for Constant AoA in MSC ADAMS.

The next form of applying lift is through a moment at the flapping hinge, applying a 'Body Fixed' torque that varies with the flapping angular velocity ($\dot{\beta}$) of the blade as shown

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = M_\beta. \quad (2.4)$$

where

$$M_\beta = K * (\theta - \dot{\beta} / \Omega). \quad (2.5)$$

The pitch is assumed to be constant for this case of simulation. The following parameters were kept consistent with both simulation and analytical formulation:

Parameter	Value	Units
K	25	Nm
θ	0.5236	rad

Table 2.2: Parameters for Moment due to Lift with Constant AoA- Basic Flap Model

The results match for the simulation and analytical solution, with respect to both the amplitude and frequency of oscillations. They both oscillate about (i.e. steady state value)

$$\beta = K * \theta / (I_x + m * d^2) \quad (2.6)$$

as expected.

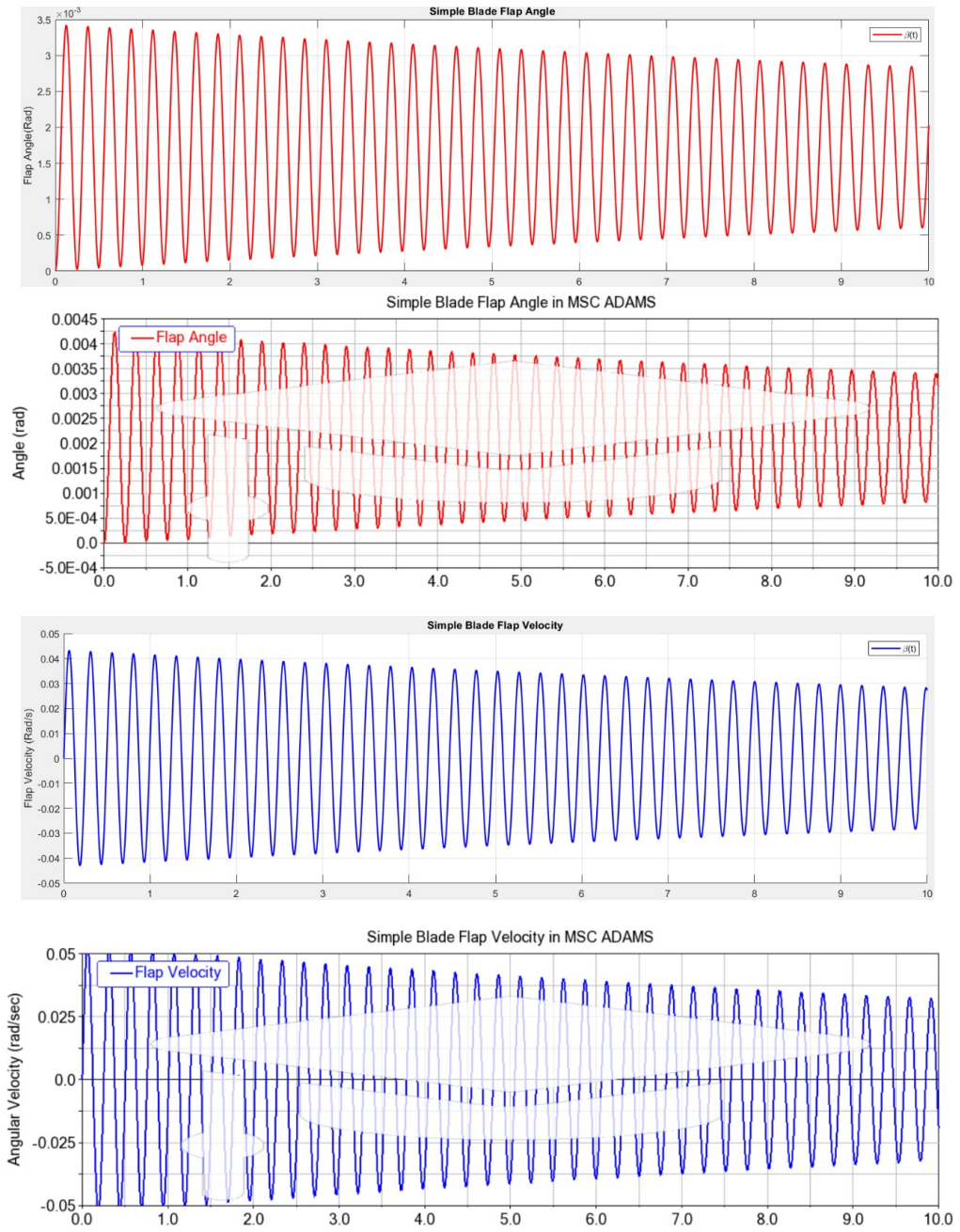


Figure 2.5: Comparison between flap angle and velocity from MATLAB and MSC ADAMS simulations.

2.3 Moment due to Lift with Varying AoA

The final validation required is for varying pitch or AoA. We provide a cyclic/harmonic control pitch through the moment on the hinge. The expected flap angle outcome would also have the same harmonics.

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = M_\beta. \quad (2.7)$$

where

$$M_\beta = K * (\theta + \theta_c \cos(w_c t) - \dot{\beta} / \Omega). \quad (2.8)$$

The steady state solution must be in the form of:

$$\beta = \beta_0 + \beta_c \cos(w_c t) \quad (2.9)$$

(when transience is lost). Where

$$\beta_0 = K * \theta / (I_x + m * d^2) \quad (2.10)$$

The relation between β_c and θ_c , and verification of amplitude and frequencies at steady state can be done by FFTs or through the Trapezoidal Method for Fourier Series. This is explained in more detail in Chapter 3.

The following parameters were kept consistent with both simulation and analytical formulation:

Parameter	Value	Units
K	25	Nm
θ	0.5236	rad
θ_c	5	rad
w_c	6	rad/s

Table 2.3: Parameters for Moment due to Lift with Controlled Varying AoA- Basic Flap Model

The results match for the simulation and analytical solution perfectly, with respect to both the amplitude and frequency of oscillations. They both oscillate about (i.e. steady state value)

$$\beta = K * \theta / (I_x + m * d^2) \quad (2.11)$$

as expected.

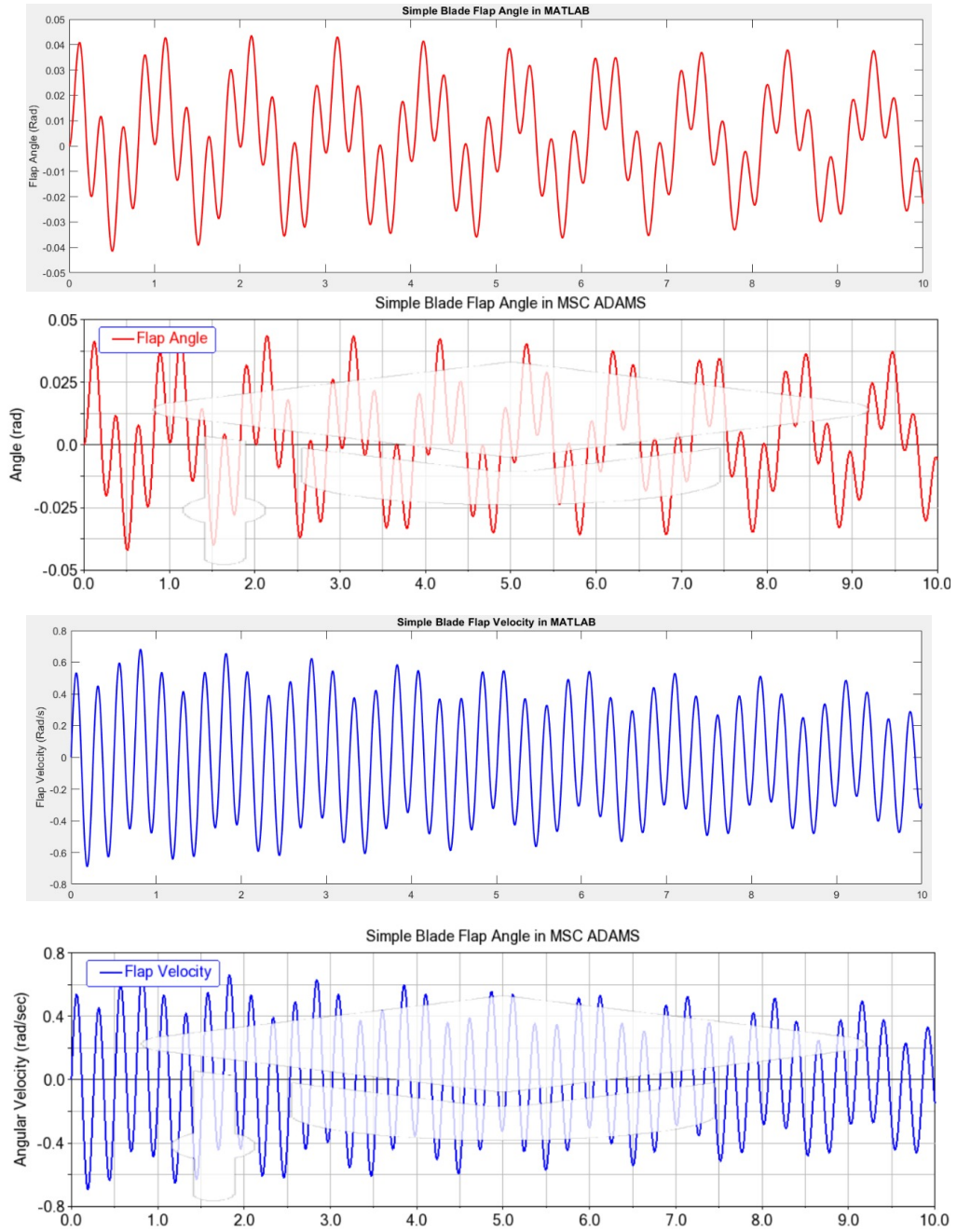


Figure 2.6: Comparison between flap angle and velocity from MATLAB and MSC ADAMS simulations.

CHAPTER 3

MODELING AND SIMULATION: ROTOR AND SWASHPLATE ASSEMBLY

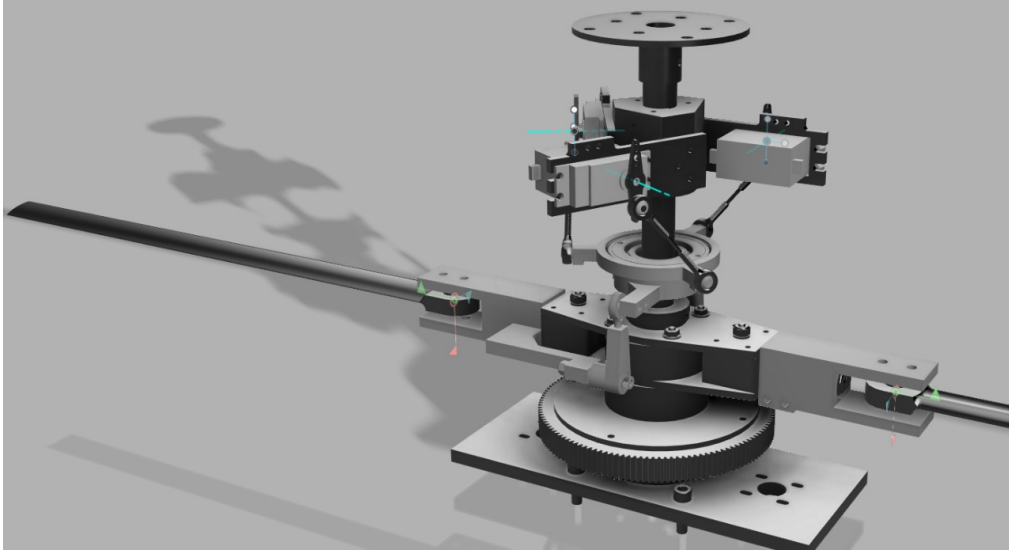


Figure 3.1: Rotor Swashplate Assembly modeled in Autodesk Fusion

To enable simulation of realistic helicopter rotor behavior, a detailed CAD model was developed on Autodesk Fusion 360 with specific design provisions for servo-driven pitch (angle of attack) control and mechanical hinges for flapping and feathering motions. The rotor assembly comprises a fixed base representing the fuselage, a driven gear interfacing with the motor to impart rotational motion, and a rotating hub that connects to the blades via revolute joints allowing pitch for blades. Transmission of pitch input is done via the servos through the tie-rods to the lower swashplate, tilting or translating the upper swashplate- and lifting the pitch links.

The swashplate assembly is designed with two concentric components: a stationary lower swashplate and a rotating upper swashplate. The upper swashplate is mechanically linked to the rotor blades through pitch links and transmits control inputs without rotating the lower swashplate. This is achieved via a bearing interface between the upper and lower swashplates, which allows differential motion. The swashplate assembly is mounted on a central spine and is

constrained using a combination of spherical and prismatic joints to allow the required translational and tilting motions.

Collective pitch control is implemented by equal servo inputs, translating the swashplate along the spine without tilt, thereby uniformly adjusting the pitch of all blades. Cyclic pitch control is achieved by differing servo inputs, tilting the swashplate, resulting in azimuth-dependent pitch variation across the rotor disk and enabling directional control of the thrust vector.

The following parameters were kept consistent with the simulation and analytical formulation:

Parameter	Value	Units
Ω (Motor speed)	25	rad/s
L (Lift)	50	N
m (Mass)	0.287286	Kg
R (Length of blade)	0.377	m
d (Eccentricity of rotating axis with flap hinge)	0.1355	m
I_x (Inertia about flap hinge x)	10	Nm^2
I_z (Inertia about flap hinge z)	10	Nm^2

Table 3.1: Parameters for Rotor Swashplate Model

The corresponding model, shown in Fig. 3.1, was imported into MSC ADAMS, where aerodynamic loads were applied following the same structured, three-stage verification process to ensure consistency with theoretical predictions. At this stage, it is sufficient to examine the moment induced by lift, using the same foundational principles established in the Simple Flap Model.

3.1 Moment due to Lift with Constant AoA

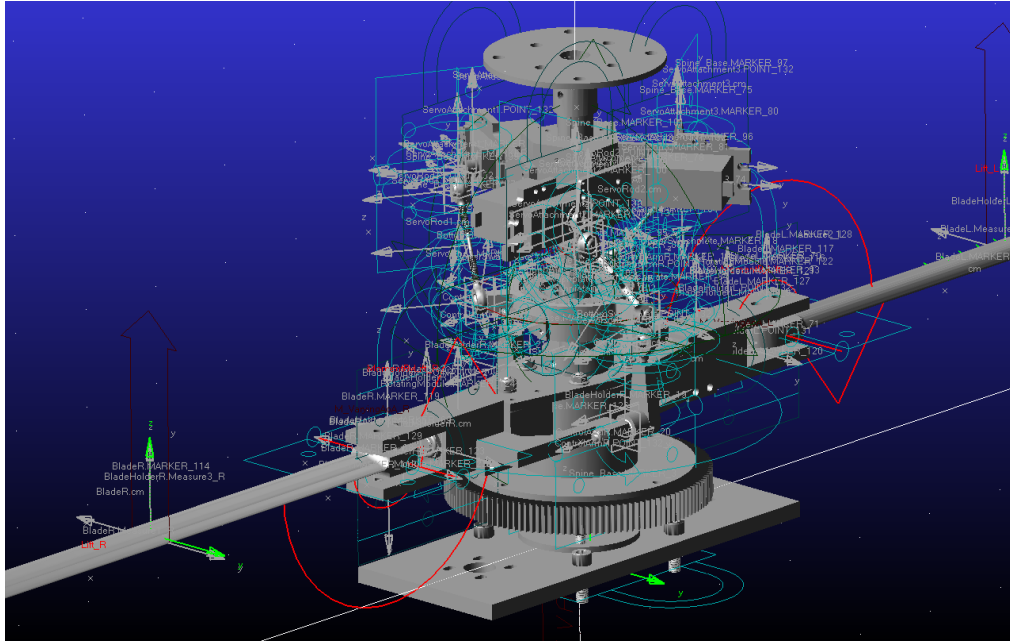


Figure 3.2: Rotor Swashplate Assembly with respective Markers.

Aerodynamic force is simulated through a moment at the flapping hinge of both blades, applying a 'Body Fixed' torque that varies with the flapping angular velocity ($\dot{\beta}$) of each blade as shown

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = M_\beta. \quad (3.1)$$

where

$$M_\beta = K * (\theta - \dot{\beta}/\Omega). \quad (3.2)$$

The pitch is constant for this case of simulation (servos are given no input). The motor drives the rotor, and the blades flap while trying to balance the lift moment and centrifugal force. The following parameters were kept consistent with both simulation and analytical formulation:

Parameter	Value	Units
K	25	Nm
θ	0.5236	rad

Table 3.2: Parameters for Moment due to Lift with Constant AoA- Rotor Swash-plate Assembly

The results of flap angle match for the simulation and analytical solution, with respect to the steady state solution and frequency of oscillations accurately (to

the third decimal for steady state solution). The flap amplitude on the other hand is slightly larger(12% error) but does not affect the average value, i.e. steady state value:

$$\beta = K * \theta / (I_x + m * d^2) \quad (3.3)$$

as expected.

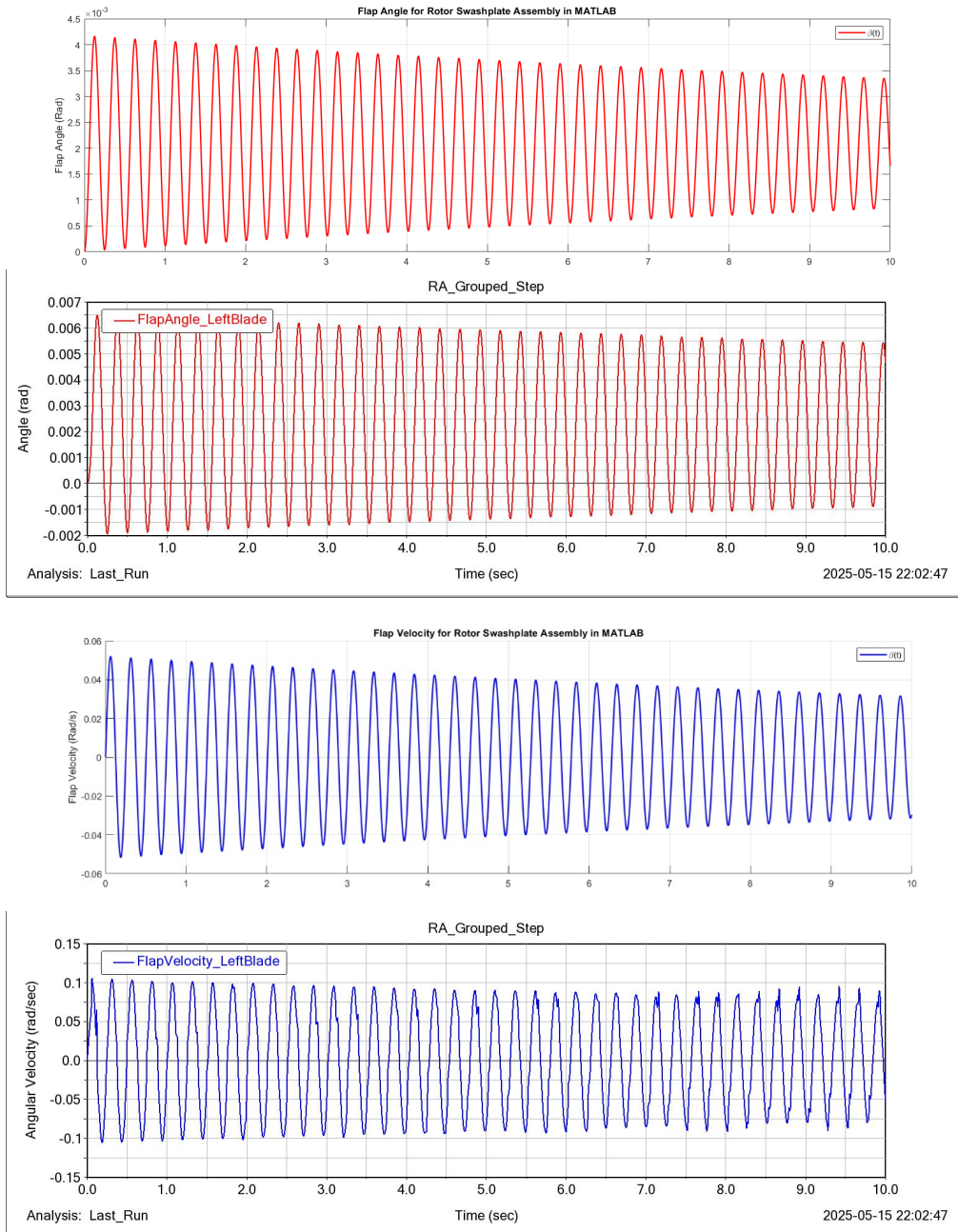


Figure 3.3: Comparison between flap angle and velocity from MATLAB and MSC ADAMS simulations.

3.2 Moment due to Lift with Varying AoA

The final validation required to ensure that the model built is accurate is to test for varying pitch or AoA. We provide a cyclic/harmonic control pitch and expect the flap angle outcome to have the same harmonics.

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = M_\beta. \quad (3.4)$$

where

$$M_\beta = K * (\theta + \theta_c \cos(w_c t) - \dot{\beta} / \Omega). \quad (3.5)$$

The steady state solution must be in the form of:

$$\beta = \beta_0 + \beta_c \cos(w_c t) \quad (3.6)$$

(when transience is lost). Where

$$\beta_0 = K * \theta / (I_x + m * d^2) \quad (3.7)$$

The following parameters were kept consistent with both simulation and analytical formulation:

Parameter	Value	Units
K	25	Nm
θ	0.5236	rad
θ_c	5	rad
w_c	6	rad/s

Table 3.3: Parameters for Moment due to Lift with Controlled Varying AoA-Basic Flap Model

The relation between β_c and θ_c analytically can be found from considering the ODE:

$$I_x \ddot{\beta} + (K/\Omega) \dot{\beta} + (I_x + m * d^2) \Omega^2 \beta = K * (\theta + \theta_c \cos(w_c t)) \quad (3.8)$$

Consider the second-order linear differential equation:

$$\ddot{\beta} + (2\zeta\omega_n)\dot{\beta} + \omega_n^2\beta = A_0 * \omega_n^2 * \exp i w_c t \quad (3.9)$$

Then

$$\beta_p = |G(i\omega t)| A_0 e^{i(\omega t - \psi)} \quad (3.10)$$

where the magnitude of the frequency response function is

$$|G(i\omega)| = \frac{1}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} = \frac{1}{\omega_n^2 \sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \quad (3.11)$$

and the phase lag ψ is given by:

$$\psi = \tan^{-1} \left(\frac{2\eta r}{1 - r^2} \right) \quad (3.12)$$

The non-dimensional frequency ratio:

$$r = \frac{\omega}{\omega_n}$$

The solution to this ODE is

$$\beta = \beta_p + \beta_c \quad (3.13)$$

Where β_c which is dependent on ω_n dies out eventually, leaving only β_c in the steady state solution. On comparing Eq 3.8 and 3.9, we get Then at steady state,

Parameter	Value	Units
ζ	0.0019974	NA
A_0	0.0199691	N.Kg/m
ω_n	25.0066	rad/s
r	0.239937	NA
β_p Amplitude = $ G(i\omega_c) * A_0$	0.021189	rad

Table 3.4: Coefficients of Harmonic Forced Second Order Linear ODE

we expect:

$$\beta_0 = K * \theta / (I_x + m * d^2) = 0.0021 \quad (3.14)$$

$$\beta_c = |G(i\omega_c)| * A_0 = 0.021189 \quad (3.15)$$

$$\beta = 0.0021 + 0.021189 \cos(\omega_c t) \quad (3.16)$$

This aligns with the Fourier Series applied on the MATLAB solution as well an FFT written for MATLAB for the analytical solution, at steady state:

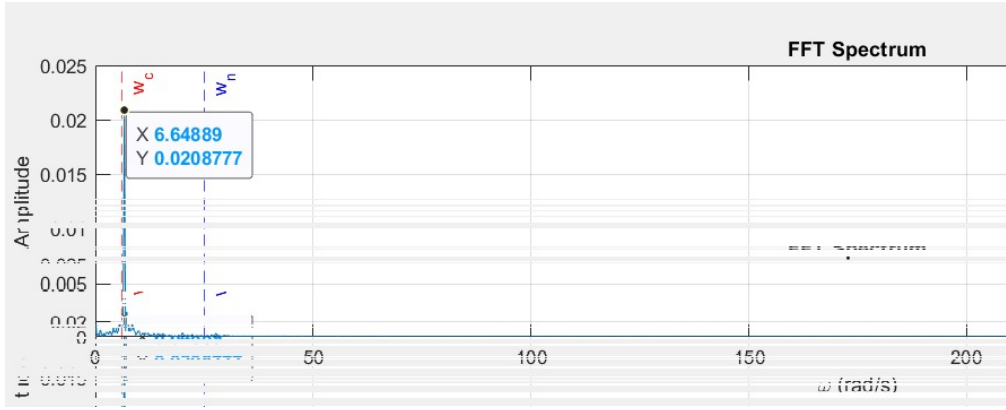


Figure 3.4: FFT of Analytical Solution on MATLAB.

The peak for the FFT occurs at $\omega_c = 6 \text{ rad/s}$ as expected (there is slight spectral leakage and transience for this data) and the practical amplitude is close to 0.021 as derived in Table 3.4.

Similarly an FFT is done for a steady state window of the Flap Angle signal from MSC ADAMS. There is still some transience left in this window and hence a peak at the natural frequency as expected for the transient solution, at $\omega_n = 25 \text{ rad/s}$ is observed as well. The peak at $\omega_c = 6 \text{ rad/s}$ is observed accurately and at the same amplitude of 0.021 (for complete steady state, without any transience).

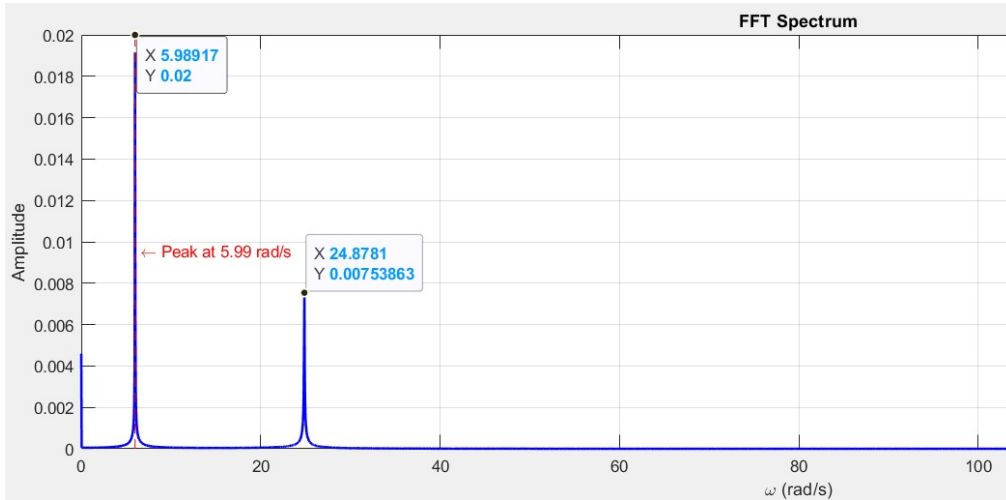


Figure 3.5: FFT of Solution on ADAMS through MATLAB.

The results for the simulation and analytical solution match perfectly with respect to both the amplitude and frequency of oscillations. They both oscillate about $\beta_0 = 0.0021$ at steady state as expected.

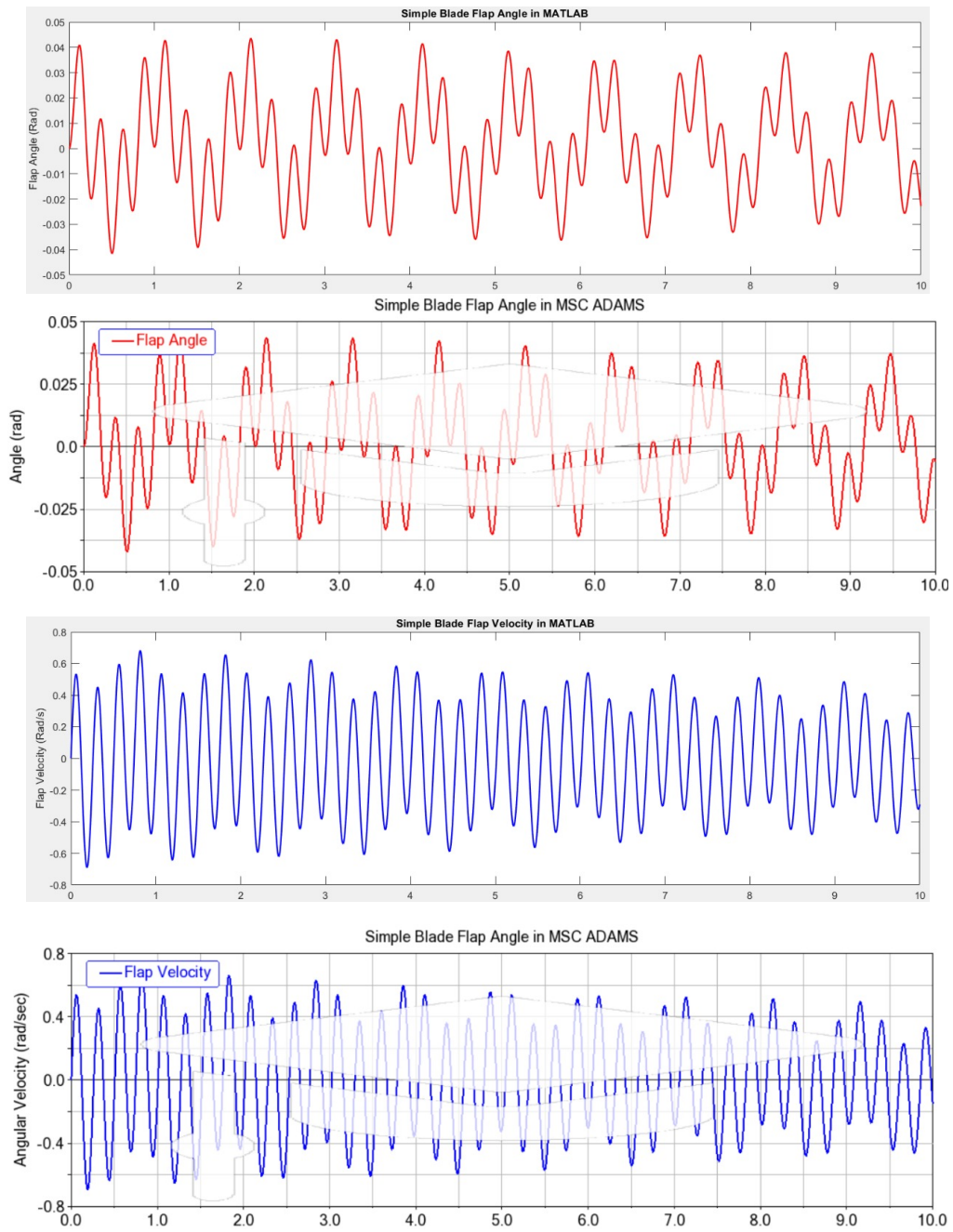


Figure 3.6: Comparison of flap angle and velocity from MATLAB and MSC ADAMS simulations.

CHAPTER 4

RESULTS AND DISCUSSION

The goal of this thesis is to study and model a rotor and swashplate assembly capable of accurately simulating blade degrees of freedom in response to control inputs and varying aerodynamic loads. In Chapter 3, Section 3.2, a harmonic pitch input is applied, and the resulting blade flapping response is shown to match the analytically predicted behavior. This successful validation establishes the model's accuracy, allowing confidence in its response to any arbitrary pitch input

The validation process began with a simplified model of a rotor blade constrained to a single degree of freedom (flapping), progressing through increasingly realistic aerodynamic forcing methods:

4.0.1 Blade Flapping Validation with Varying Aerodynamic Inputs

Point Lift Application (Chapter 2.1, Fig. 2.2, 2.3) A space-fixed vertical force representing lift was applied at the center of mass of the blade. The resulting flap angle and angular velocity matched the analytical solution well, with less than 10% deviation in amplitude and excellent agreement in frequency. Both systems oscillated about the theoretical equilibrium value of

$$\beta = \frac{M_\beta}{I_x + md^2}$$

which was clearly demonstrated in Fig. 2.3.

Constant AoA Moment Input (Chapter 2.2, Fig. 2.4, 2.5) A body-fixed torque was applied based on a constant pitch angle. The simulation confirmed

the correct dynamic response and equilibrium flap angle. The steady-state solution matched the analytical result from MATLAB, as seen in Fig. 2.5, again validating the model accurately.

Harmonic AoA Moment Input (Chapter 2.3, Fig. 2.6) A time-varying pitch angle (cyclic pitch control) was modeled via a harmonic cosine input. The flap angle $\beta(t)$ exhibited the expected oscillatory behavior:

$$\beta(t) = \beta_0 + \beta_c \cos(\omega_c t)$$

with steady-state amplitude and frequency matching the input forcing. The oscillation amplitude and frequency matched precisely with MATLAB results (see Fig. 2.6), confirming the integrity of the simulation setup.

4.0.2 Full Rotor–Swashplate Assembly Validation

Constant AoA Case (Chapter 3.1, Fig. 3.3) A realistic multibody model of the rotor and swashplate was constructed in Fusion 360 and imported into ADAMS. When a constant pitch angle was applied, the blades flapped and reached steady-state values that matched the analytical solution closely. The flap angle error was under 12% in amplitude, which drastically decreased with decrease in time step, and the steady-state value matched to the third decimal (see Fig. 3.3). The error seems to arrive due to the employment of different solvers in MATLAB and ADAMS.

Harmonic AoA Case (Chapter 3.2, Fig. 3.6) With a sinusoidal pitch input, the flap angle showed a steady-state oscillation about the expected equilibrium. A Fourier analysis, confirmed with both Fourier series coefficients and FFT (Fig. 3.4 and 3.5) revealed a dominant frequency at $\omega_c = 6$ rad/s, exactly matching the forcing frequency. The amplitude observed from the simulation was $\beta_c = 0.021$ rad, which perfectly agreed with the analytically predicted amplitude (Table 3.4). Residual transient frequencies, like the natural flapping frequency $\omega_n = 25$ rad/s, were minimal in the steady-state window.

4.0.3 Validation Summary

Across both simplified and full-assembly models:

- Amplitude and frequency of flap angle response matched theoretical predictions.
- Time-domain simulations in ADAMS closely followed MATLAB solutions.
- Frequency-domain validation using FFT and Fourier series matched analytical solutions, confirming model accuracy even for harmonic forcing.

These results establish high confidence in the rotor-swashplate assembly model built in ADAMS and its ability to capture realistic flapping behavior under control-induced aerodynamic loads.

4.0.4 Future Scope

With a validated and accurate model of the helicopter rotor system the model's use cases can be extended to several other major focuses:

- **Flexible Blades:** Simulating blade flexibility is important for realistic helicopter blade dynamics. Some rotors also have "virtual" hinges based on this flexibility. Insights into aeroelastic effects and blade stability under different flight regimes are important.
- **Cyclic and Collective Pitch Inputs:** The current model already supports steady and harmonic collective inputs. Extending this to simulate realistic cyclic pitch control enables full control analysis, critical for maneuvering and forward flight simulation.
- **Flexible Multibody Dynamics:** Transitioning from rigid body models to flexible bodies for the swashplate or blade holders would allow a more refined study of structural vibrations, resonance, and fatigue.

Ultimately, the developed framework lays a strong foundation for advanced aero-mechanical studies in rotorcraft dynamics, with potential applications in both research and rotorcraft design optimization.

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